

CEE 398

Linear Potentiometers for Structural Engineering

Ping Gu
University of Illinois

Outline:

- Introduction
- Principle
- Features
- wound-wire Pot and thin-film Pot
- Comparison with LVDT
- Examples

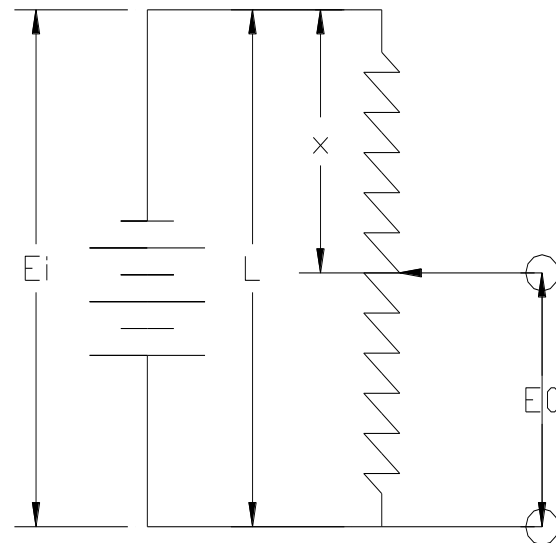
Introduction:

- Name: Linear Potentiometers (Pot)
- Usage: Measure displacement
- instruments for Displacement Measurements:
 - Optical:
 - **Laser tracker**
 - **Video camera and digital computer motion analysis**
 - Mechanical: the most commonly used is dial gage
 - **individual readings by eye**
 - **0.025~0.0025mm division**
 - **6.25~150mm range**
 - Acoustic: ultrasonic sound wave
 - Electrical: (Displacement transducers; Continuous recording)
 - **resistive: potentiometers**
 - **inductive: LVDT;VIP sensors**
 - **capacitive:**

Principle:

- Constant input E_i @ends of fixed resistance
- Movable slider (wiper) contacting the resistor
- Straight wire resistors are not feasible. Resistance of a short length is too low, requires excessive power on voltage source.
- Based on the resistor material, POT can be classified into two types: wound-wire Pot and thin-film Pot.

$$E_o = \frac{x}{L} E_i$$



Features:

- stroke: up to 1.5 meters
- resistance range 10~1,000,000 Ohms.
- the signal conditioning equipment must have high input impedance
- inertia and friction limit frequency response, precludes its use for dynamic testing.
- contact wears rapidly in continuous service. Wiper made of precious metal.
- a change in E_i or the shunting effect of a low resistance in the output circuit changes the current, becoming nonlinear

wound-wire Pot and thin-film Pot

■ wound-wire Pot

- resolution: wiper jumps from one wire to another, resolution is limited to L/n (Length of the coil/ number of turns), so it is used to measure large displacements (10mm or more). Common resolution 0.05~1% , with 0.05% by many turns of very fine wire.
- Electronic noise due to the movement of wiper from one wire to another. Cleaning and lubrication of the coil.
- Life is about 1 million cycles

■ Thin-film Pot (thin films with controlled resistivity)

- high resolution (e.g. conductive plastic films 50-100 ohm/mm resolution 0.001 mm)
- lower noise
- longer life (>10 million cycles). (e.g. conductive plastic films 100 million cycles)

Comparison with LVDT:

■ Disadvantages:

- can only be used for static or quasi-static measurements due to friction and inertia
- resolution
- reliability problem

■ On the other side, LVDT

- frictionless measurement; non-contact
- suitable for dynamic testing
- infinite resolution
- long life

■ Advantages:

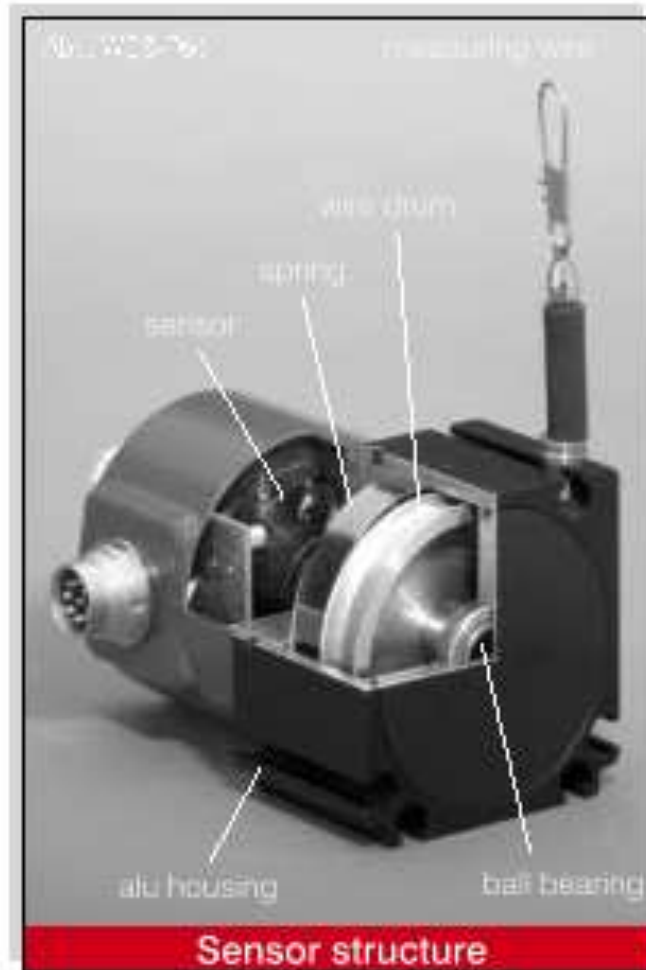
- simplicity of operation
- inexpensive
- accurate

Example One:

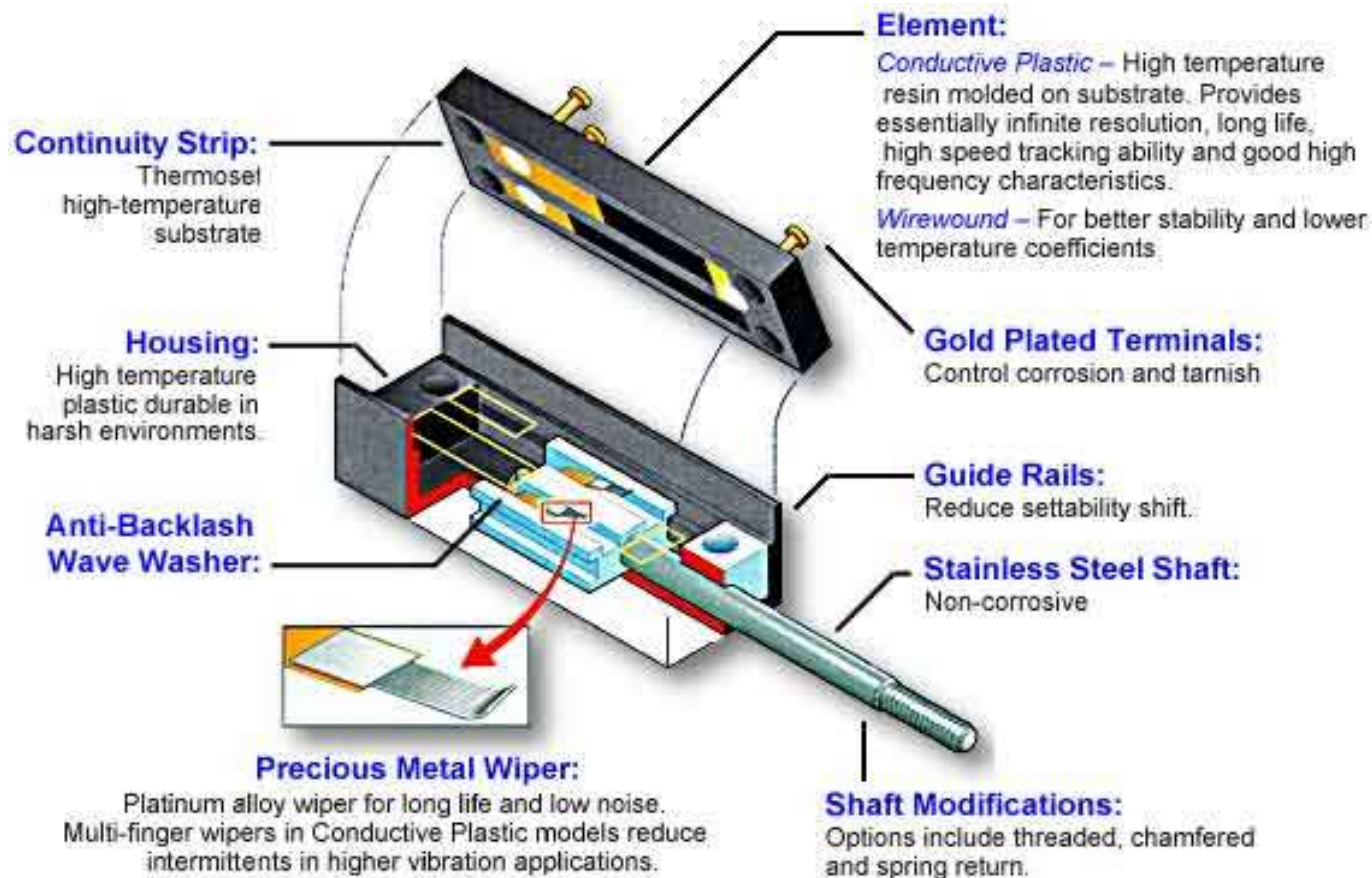


- this one: Celeasco PT1DC
- stroke: 2 inches (in this series 2~50 inches)
- input $E_i = 14.5 \sim 40$ V DC
- output signal $0 \sim 10$ V DC
- plastic-hybrid precision pot
- resolution essentially infinite
- measuring cable 0.019 in diameter (0.48 mm)
- cycle life 2.5×10^6
- output signal about 4000 ohms, output impedance 1000 ohms, max. output load 5000 ohms

Example Two:



Example Three:



Conclusion:

- Introduction
- Principle
- Features
- wound-wire Pot and thin-film Pot
- Comparison with LVDT
- Examples